2013 MTTC Pennsylvania Regional Skills Contest



April 22-25, 2013

Problem Statement

This morning a foreman and three crew members entered the section to continue the mining process.

Just a few minutes ago, an explosion was heard and the fan stopped. We haven't heard from the foreman and crew and the tracking system shows that they have not left the section.

Also after the explosion, the drift openings going into the mountain became blocked by water and unsafe roof. A mine examiner was able to make the no.2 drift opening and found no explosive gases, but he was blocked from further travel by unsafe roof. So a decision was made to enter the mine by using the escape elevator. You are located outside beside the escape elevator at this time. This elevator will drop you and your team down into the no.2 entry inby the last open crosscut. The mine was preparing for a 90 degree turn after hitting bad top, water and dangerous amounts of methane while driving a northerly direction.

Intake air was being pulled into the no.1 entry drift opening and across the last open to the no.3 entry air shaft with an exhaust fan on top of the shaft. The fan is off at this time. The escape elevator shaft in no. 2 entry can also be used as an intake if needed.

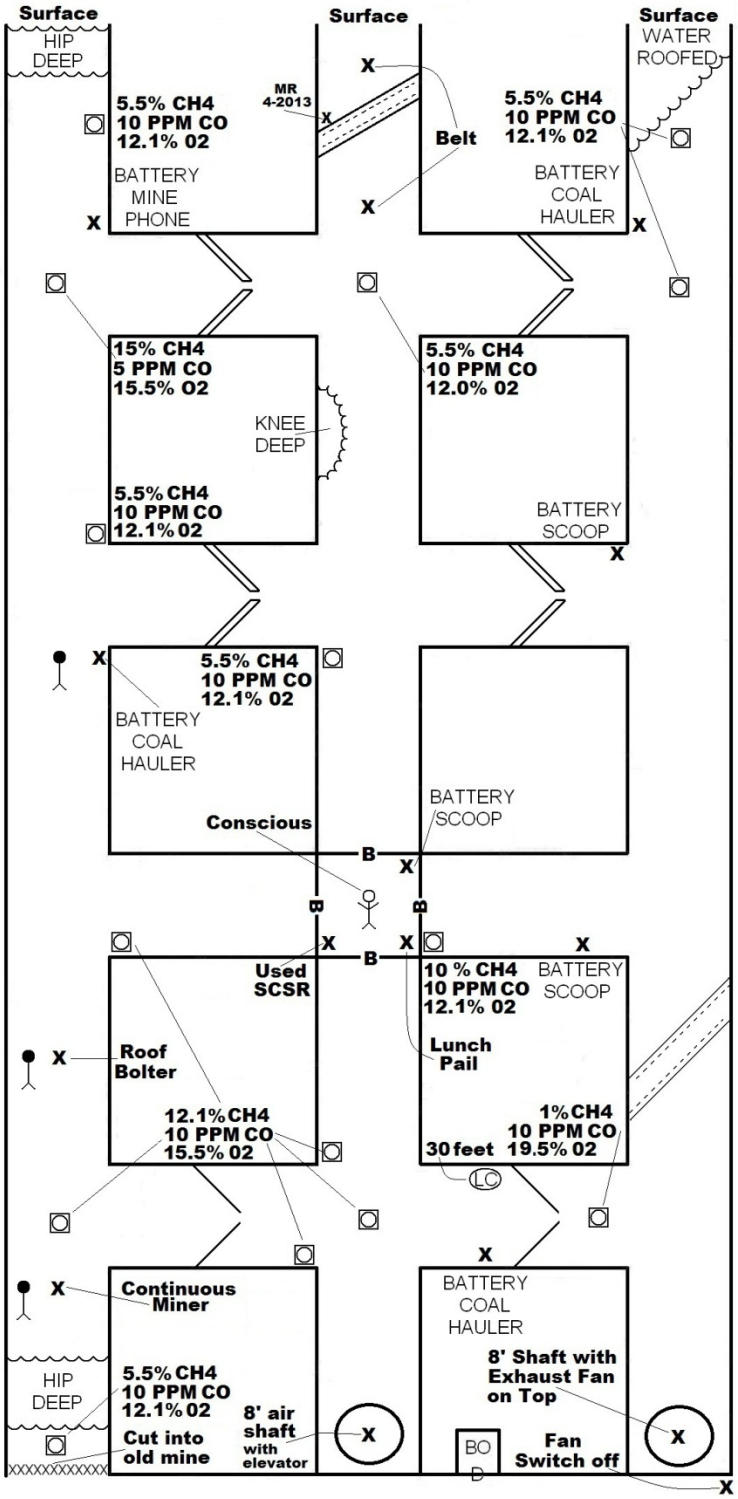
This mine was mining close to an old mine, but we believe our distance from the old mine is safe.

We have never had a problem before with our contract engineering company making mistakes.

The mine map should be very close to correct. You will receive two copies of the section map when you are ready to start. The Briefing Officer's map is identified.

Back up Mine Rescue teams are ready and available if you need assistance. You can begin when you are ready!

2013 MTTC PA Skills



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Written Instructions

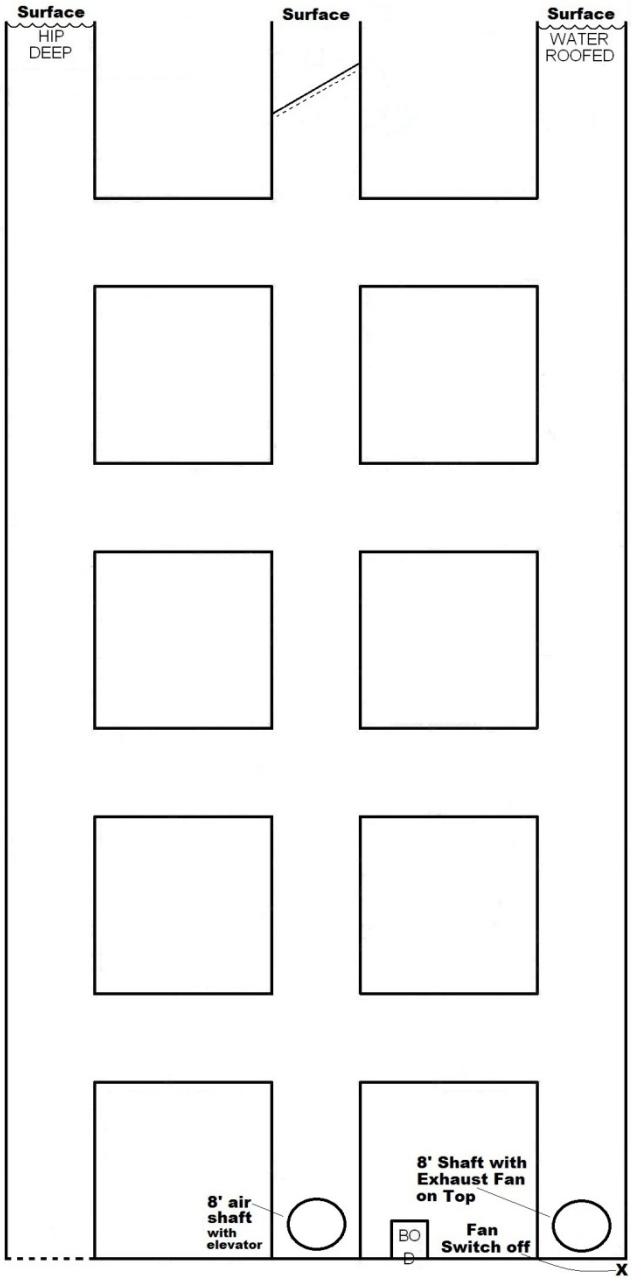
BRING ALL SUVIVORS TO THE FRESH AIR BASE

THE EXHAUST FAN CANNOT BE STOPPED OR STALLED AFTER STARTING. THE EXHAUST FAN CAN NEVER BE REVERSED

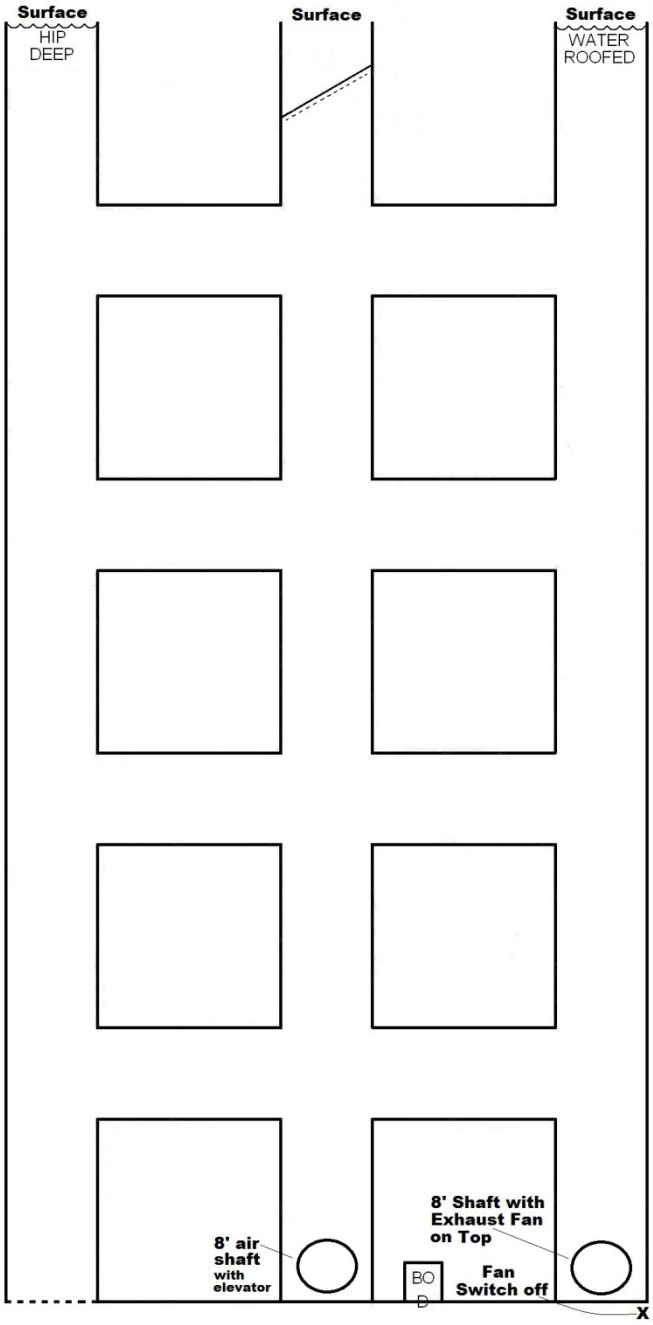
THE BRIEFING OFFICER HAS BEEN ISOLATED AND MUST REMAIN IN A RESPIRABLE ATMOSPHERE

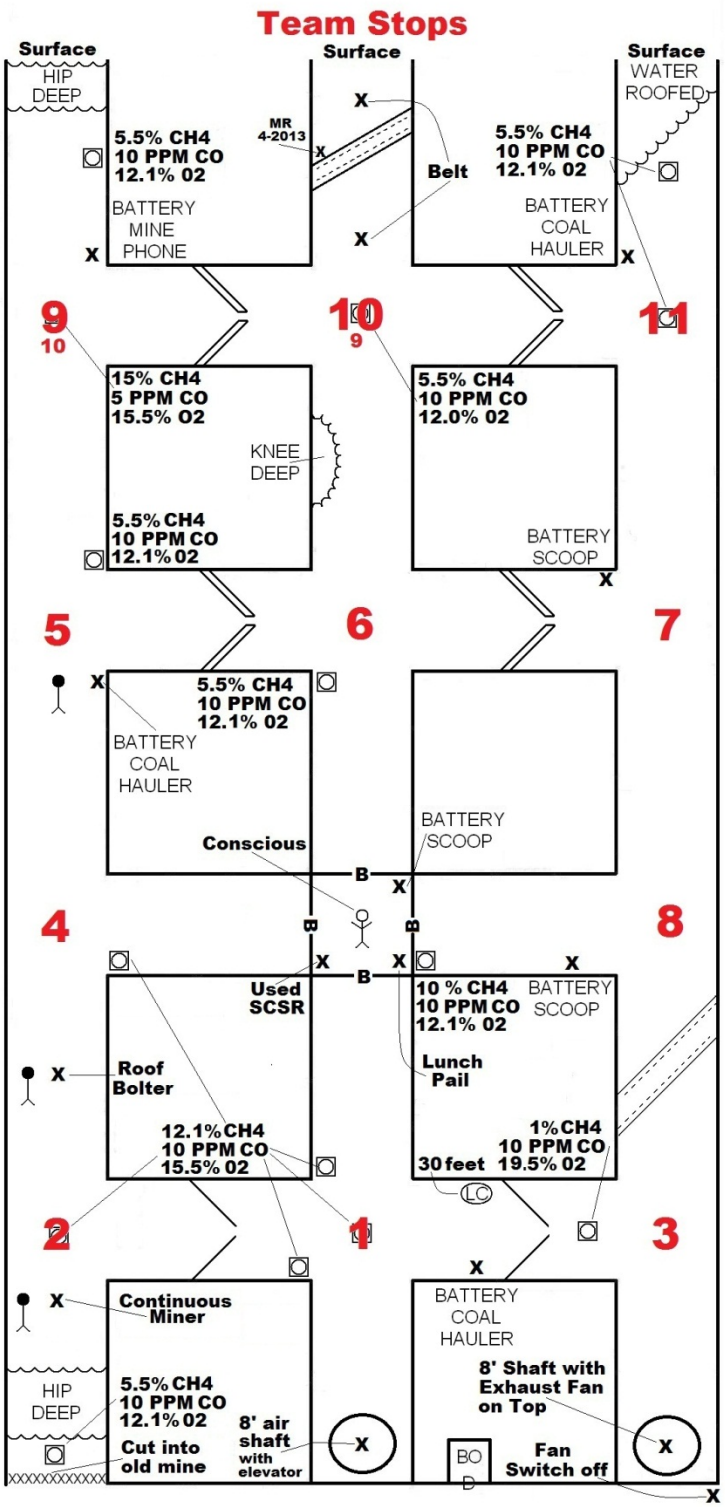
UP OR DOWN ON THE ELEVATOR WILL HAPPEN IMMEDIATELY WHEN THE CAPTAIN TELLS THE CHAIRMAN OF JUDGES "TEAM IS READY TO GO UP / DOWN"

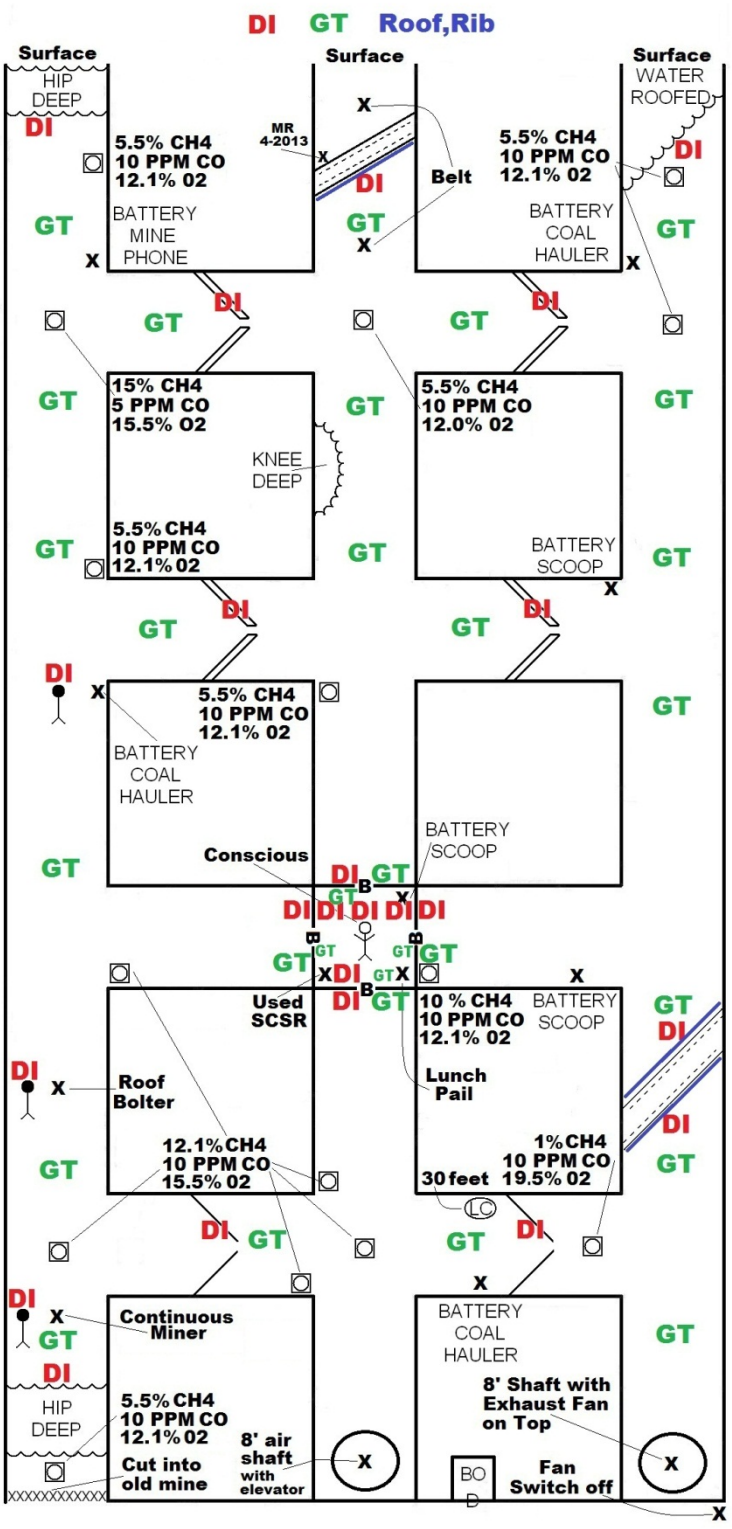
BO Map

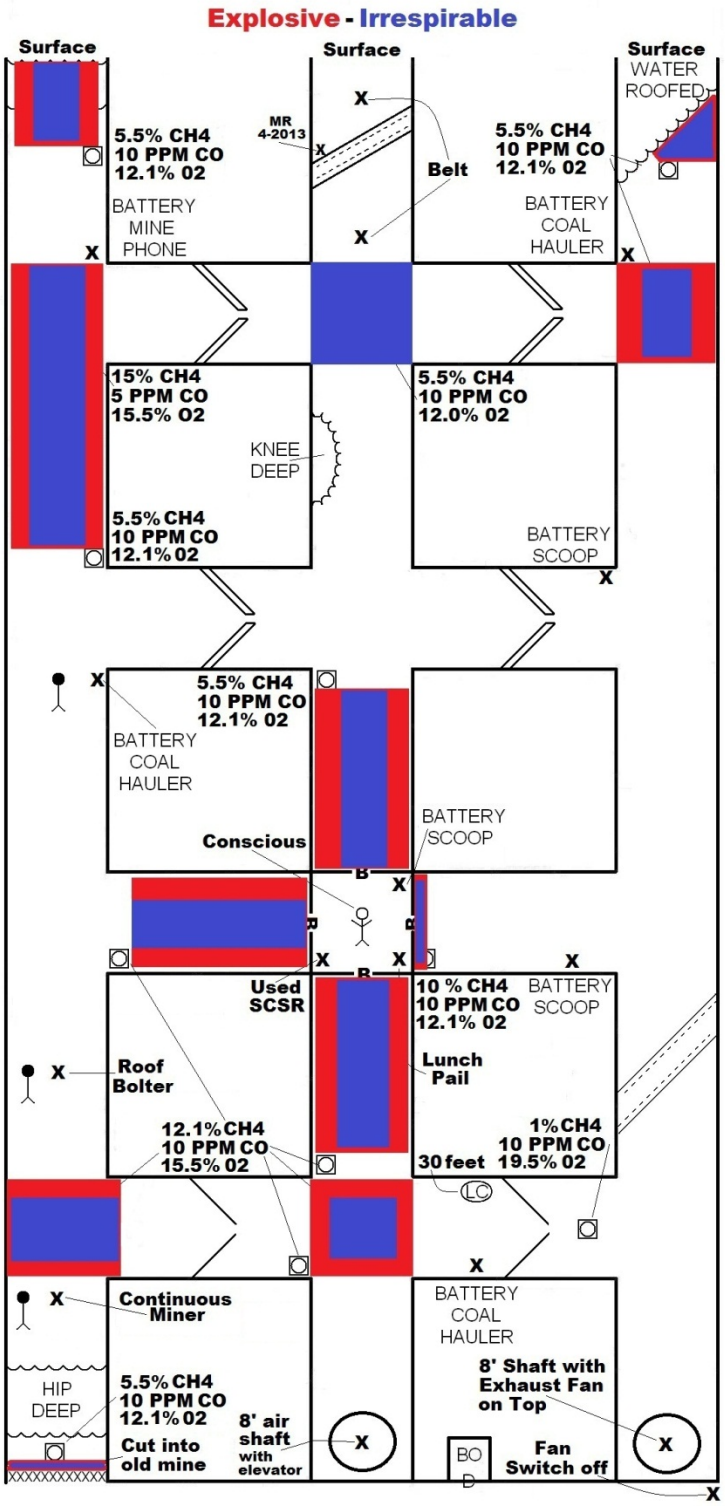

Surface
HIP
DEEP
Surface
Surface
WATER
ROOFED
8' air
shaft
with
elevator
8' Shaft with
Exhaust Fan
on Top
BO
D
Fan
Switch off
X

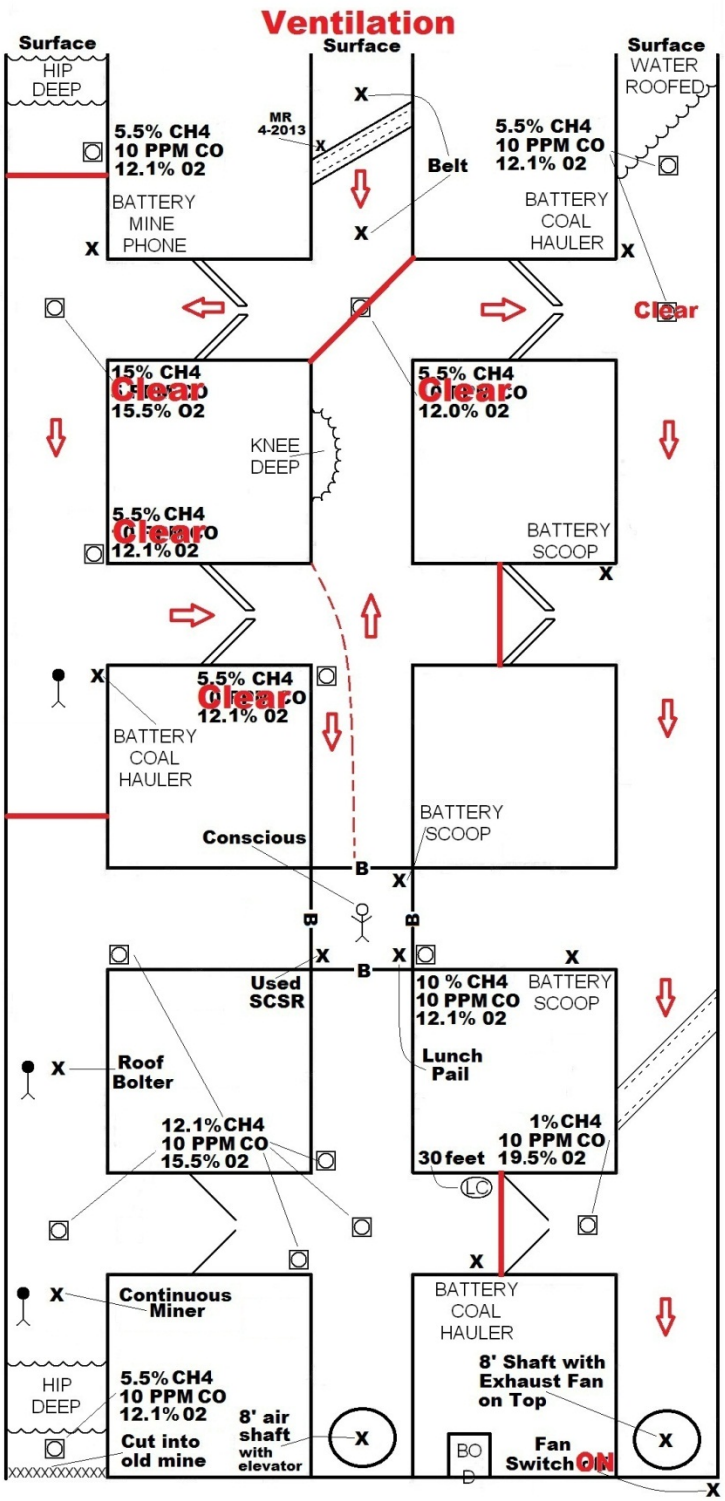
Team Map


Surface
HIP
DEEP
Surface
Surface
WATER
ROOFED
8' air
shaft
with
elevator
8' Shaft with
Exhaust Fan
on Top
BO
D
Fan
Switch off
X


Team Stops
Surface
HIP
DEEP
5.5% CH4
10 PPM CO
12.1% 02
BATTERY
MINE
PHONE
MR
4-2013
Surface
Belt
Surface
WATER
ROOFED
5.5% CH4
10 PPM CO
12.1% 02
BATTERY
COAL
HAULER
9
10
15% CH4
5 PPM CO
15.5% O2
KNEE
DEEP
5.5% CH4
10 PPM CO
12.1% 02
10
9
5.5% CH4
10 PPM CO
12.0% 02
BATTERY
SCOOP
11
5
6
7
5.5% CH4
10 PPM CO
12.1% 02
BATTERY
COAL
HAULER
Conscious
BATTERY
SCOOP
B
4
8
Used
SCSR
10 % CH4
10 PPM CO
12.1% 02
BATTERY
SCOOP
Roof
Bolter
Lunch
Pail
12.1% CH4
10 PPM CO
15.5% 02
1% CH4
10 PPM CO
19.5% 02
30 feet
LC
2
1
3
Continuous
Miner
BATTERY
COAL
HAULER
8' Shaft with
Exhaust Fan
on Top
HIP
DEEP
5.5% CH4
10 PPM CO
12.1% 02
Cut into
old mine
8' air
shaft
with
elevator
BO
D
Fan
Switch off


DI GT Roof,Rib
Surface
HIP
DEEP
DI
5.5% CH4
10 PPM CO
12.1% 02
BATTERY
MINE
PHONE
MR
4-2013
Surface
Belt
Surface
WATER
ROOFED
5.5% CH4
10 PPM CO
12.1% 02
BATTERY
COAL
HAULER
15% CH4
5 PPM CO
15.5% O2
KNEE
DEEP
5.5% CH4
10 PPM CO
12.1% 02
5.5% CH4
10 PPM CO
12.0% 02
BATTERY
SCOOP
5.5% CH4
10 PPM CO
12.1% 02
BATTERY
COAL
HAULER
Conscious
BATTERY
SCOOP
Used
SCSR
10 % CH4
10 PPM CO
12.1% 02
BATTERY
SCOOP
Roof
Bolter
Lunch
Pail
12.1% CH4
10 PPM CO
15.5% 02
1% CH4
10 PPM CO
19.5% 02
30 feet
LC
Continuous
Miner
BATTERY
COAL
HAULER
8' Shaft with
Exhaust Fan
on Top
HIP
DEEP
5.5% CH4
10 PPM CO
12.1% 02
Cut into
old mine
8' air
shaft
with
elevator
BO
D
Fan
Switch off

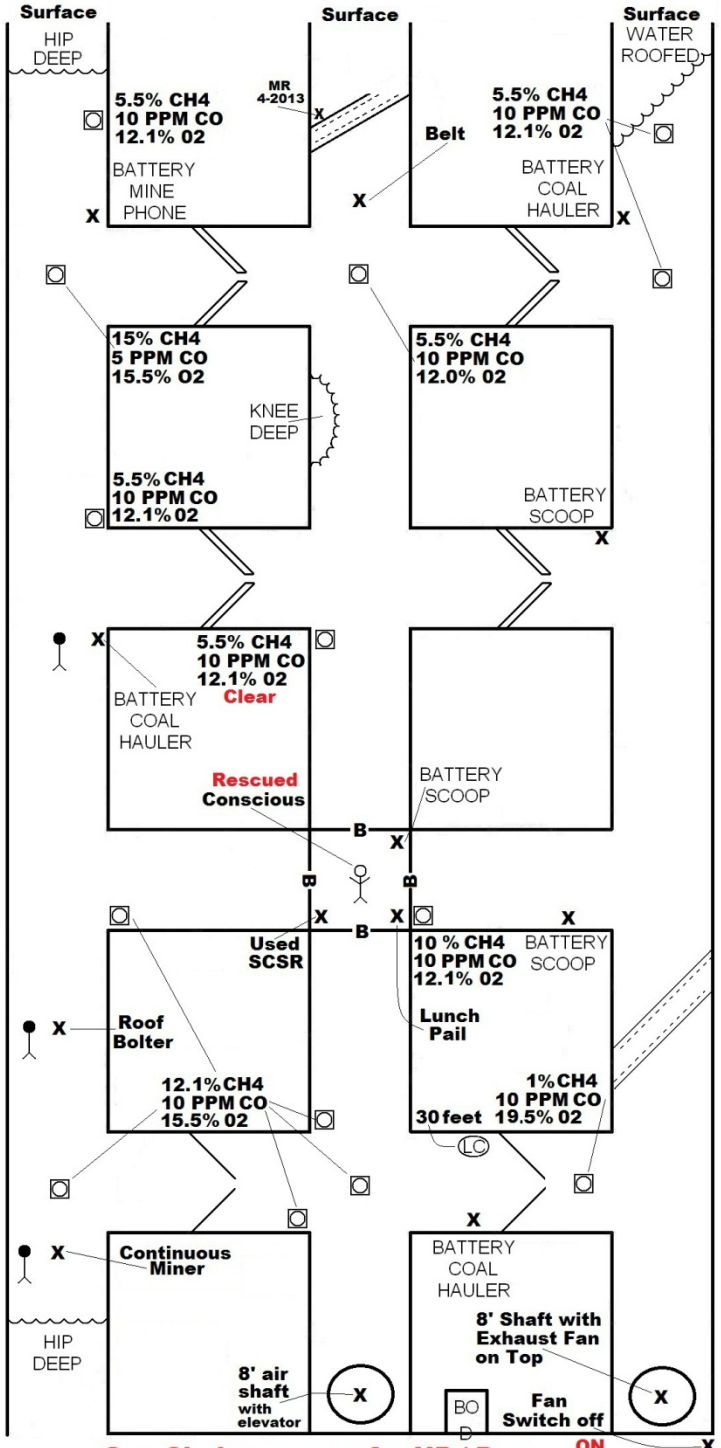

Explosive - Irrespirable
Surface
Surface
Surface
WATER
ROOFED
5.5% CH4
10 PPM CO
12.1% 02
MR
4-2013
Belt
5.5% CH4
10 PPM CO
12.1% 02
BATTERY
MINE
PHONE
BATTERY
COAL
HAULER
15% CH4
5 PPM CO
15.5% O2
KNEE
DEEP
5.5% CH4
10 PPM CO
12.1% 02
5.5% CH4
10 PPM CO
12.0% 02
BATTERY
SCOOP
5.5% CH4
10 PPM CO
12.1% 02
BATTERY
COAL
HAULER
Conscious
BATTERY
SCOOP
Used
SCSR
10 % CH4
10 PPM CO
12.1% 02
BATTERY
SCOOP
Roof
Bolter
Lunch
Pail
12.1% CH4
10 PPM CO
15.5% 02
1% CH4
10 PPM CO
19.5% 02
30 feet
LC
Continuous
Miner
BATTERY
COAL
HAULER
HIP
DEEP
5.5% CH4
10 PPM CO
12.1% 02
Cut into
old mine
8' air
shaft
with
elevator
8' Shaft with
Exhaust Fan
on Top
BO
D
Fan
Switch off


Ventilation
Surface
HIP
DEEP
5.5% CH4
10 PPM CO
12.1% 02
BATTERY
MINE
PHONE
MR
4-2013
Surface
Belt
Surface
WATER
ROOFED
5.5% CH4
10 PPM CO
12.1% 02
BATTERY
COAL
HAULER
Clear
15% CH4
Clear
15.5% O2
KNEE
DEEP
5.5% CH4
Clear
12.1% 02
5.5% CH4
Clear
12.0% 02
BATTERY
SCOOP
5.5% CH4
Clear
12.1% 02
BATTERY
COAL
HAULER
Conscious
BATTERY
SCOOP
Used
SCSR
10 % CH4
10 PPM CO
12.1% 02
BATTERY
SCOOP
Roof
Bolter
Lunch
Pail
12.1% CH4
10 PPM CO
15.5% 02
1% CH4
10 PPM CO
30 feet 19.5% 02
LC
Continuous
Miner
BATTERY
COAL
HAULER
HIP
DEEP
5.5% CH4
10 PPM CO
12.1% 02
Cut into
old mine
8' air
shaft
with
elevator
8' Shaft with
Exhaust Fan
on Top
BO
D
Fan
Switch ON

PATIENT STATEMENT FROM BEHIND ALL 4 BARRICADES

HELP! I AM SURROUNDED BY AIRTIGHT BARRICADES!

Key for Map Judges



See Chairman map for UP / Down stoppings left in the mine

Keys to Problem

- **Team must not move battery mine phone in no.1 entry. Explosive mixture is in the intersection**
- **Team must keep explosive mixtures from being moved over all batteries and unexplored areas before turning on the fan**
- **Fan cannot be stopped, stalled or reversed after starting**
- **Team must keep caved area in no.1 entry and elevator shaft in no. 2 entry blocked after fan has been started. Team can airlock back to elevator**
- **Team must protect patient from low O2 while retreating back to FAB**
- **Team must explore entire mine**